

THE STORY OF ASTRONOMY

ONE OF THE OLDEST SCIENCES, ASTRONOMY MAY HOLD THE KEY TO THE ORIGIN OF EVERYTHING

The development of astronomy has been influenced by two key factors: the invention of the telescope, which revealed previously undetectable celestial objects, and advances in mathematics, physics, chemistry, and computing, which have been crucial to explaining astronomical observations.

Early astronomy was closely linked to mythology, religion, and prognostication. Celestial observations were used to measure time, devise calendars, set the dates of religious festivals, and for astrological prediction. For millennia, it was believed that the Earth was the center of the cosmos. However, this did not fully explain the observed movements of the Moon, Sun, and planets.

MODERN ASTRONOMY

In 1543, Nicolaus Copernicus published his heliocentric model, which put the Sun at the center of the cosmos and is widely considered to mark the birth of modern astronomy. Then, after 1609, the newly invented telescope revealed a host of new astronomical objects. The 17th century also saw the establishment of the laws of planetary motion by Johannes Kepler, and an explanation



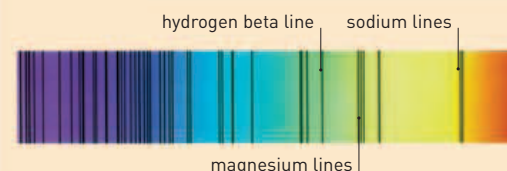
Persian astrolabe

Astrolabes show a representation of the night sky and were used until the 17th century to estimate time and for navigation.

of the gravitational force controlling that movement by Isaac Newton. In the 19th century, the distance to the Sun and nearby stars was accurately measured, spectroscopy was introduced, and advances in theoretical physics provided explanations for problems such as how stars generate their energy (by nuclear reactions in their cores). Prior to 1920, many thought the Universe consisted of only our own Milky Way Galaxy. However, Edwin Hubble measured the speed at which distant nebulae were receding, and it was realized that these nebulae were independent galaxies. Not only were the galaxies moving away, but the speed they were moving away increased with distance, implying that the Universe had a beginning, when everything was close together. It was proposed that the expansion had been caused by a massive explosion—the Big Bang. Findings from modern space astronomy have supported the Big Bang theory, but it has also been discovered that much of the Universe consists of dark matter and dark energy, the nature and origin of which are still unknown.



Newton's telescope
(front view)



THE ELECTROMAGNETIC SPECTRUM

Stars and other astronomical objects emit light and other forms of electromagnetic energy, such as X-rays and radio waves. Using spectroscopy, these electromagnetic emissions can be broken up into a spectrum of colors. A star's spectrum is crossed by dark absorption lines, each corresponding to a different chemical element. By investigating the intensity of these lines, a star's chemical composition can be discovered. Further study can also establish its temperature, relative velocity, and the pressure and density of its atmosphere.

2000 BCE

Solar and lunar calendars
The Babylonians produce the first calendar by integrating the 365.25 days of the solar year with the 29.53 days of the lunar month. Similar calendars are used in ancient Egypt.



Zodiac of Senenmut



Babylonian boundary stone

c. 90–168 CE

Ptolemy's Universe
Greek polymath Claudius Ptolemy proposes that the Earth is the center of the cosmos, a view that prevailed until the 16th century.



Ptolemy's constellations

c. 1400 BCE

Deities and the Zodiac
The ancient Egyptians produce the earliest known representation of the Zodiac, in which stars, planets, and associated deities appear. Zodiacs also appear in Babylonian artifacts.

1420

Ulugh Beg
The Persian Ulugh Beg builds an observatory in Samarkand. He measures the tilt of Earth's axis to 1/100th of a degree.



Ulugh Beg observatory

1543

The Sun-centered Universe
Nicolaus Copernicus suggests the Earth orbits the Sun and not vice versa. This demotes the Earth to being just one of the six known planets.



The Copernican Solar System

1608/1668

The first telescopes
German-born Dutch lensmaker Hans Lippershey makes the first refracting telescope in 1608. English scientist Isaac Newton makes the first reflecting telescope in 1668.

Newton's telescope

